

# Model Evaluation - Study Notes

## Data splits

Data is usually divided into a training set, a validation set and a test set. Common ratios are 70/15/15 or 80/10/10. The training set is used to fit the model, the validation set to tune hyperparameters, and the test set to estimate final performance. The test set must be used only once, at the end.

## Cross-validation

In k-fold cross-validation the data is split into k folds. The model is trained k times, each time using a different fold as the validation set, and the k scores are averaged. Typical values of k are 5 or 10. Stratified k-fold keeps the class proportions the same in every fold, which is important for classification.

## Classification metrics

Accuracy is the number of correct predictions divided by the total number of predictions. Precision is TP divided by (TP + FP): of all the examples predicted positive, how many are truly positive. Recall is TP divided by (TP + FN): of all the truly positive examples, how many the model found. The F1 score is the harmonic mean of precision and recall, computed as  $2PR / (P + R)$ . A confusion matrix shows the counts of true positives, false positives, true negatives and false negatives.

## ROC and AUC

The ROC curve plots the true positive rate against the false positive rate at different decision thresholds. The area under the curve (AUC) summarizes it in one number: 0.5 means random guessing and 1.0 means a perfect classifier.

## Imbalanced datasets

Accuracy can be very misleading when classes are imbalanced. If 99 percent of transactions are legitimate, a model that always predicts the majority class, legitimate, reaches 99 percent accuracy while catching no fraud at all. For imbalanced problems use precision, recall, F1 or the area under the precision-recall curve instead of accuracy.

## Regression metrics

Mean absolute error (MAE) is the average absolute difference between predictions and true values. Mean squared error (MSE) squares the errors, so large errors are penalized more. Root mean squared error (RMSE) is the square root of MSE and is expressed in the same units as the target. R-squared is the proportion of variance explained by the model: 1 is perfect and it can even be negative for a model worse than predicting the mean.